Revision History

Date	Version	Description
September 4, 2019	0.50	Initial version created.
September 19, 2019	0.51	Updated the Program Storage and Low Power State section.
March 13, 2020	0.52	Updated Hibernation and Retention Comparison section, Table 3, Table 5, Figure 1, 2, Table A-1, Table A-2.
August 10, 2020	0.53	Updated Overview, Hibernation and Retention Comparison, Hibernation and SoC Off Comparison sections, Table A-1.
December 2, 2020	0.54	Corrected typos.
March 29, 2021	0.55	Updated Table 3, Figure 1.
April 14, 2021	0.56	Updated format, no content change.

CONFIDENTIAL

©2019-2021, Atmosic Technologies Inc. All rights reserved. Atmosic logo is a registered trademark of Atmosic Technologies Inc. All other trademarks are the properties of their respective holders. This document is subject to change without notice.

Table of Contents

Overview 3

Purpose 4

Hibernation and Retention Comparison 4

Use Case Example 5

Hibernation and SoC Off Comparison 6

Program Storage and Low Power State 10

Developing a User-Defined Program 10

Examination of SDK's tmp1075_sensor_adv Application 12

Appendix A 13

List of Figures

Figure 1 - Hibernation and Retention Average Power vs Time Comparison
Figure 2 - Hibernation and SoC Off Average Power vs Time Comparison
Figure 3 - User-defined Cyclical Application Development Process

List of Tables

Table 1 - Power States
Table 2 - Hibernation vs. Retention Comparison
Table 3 - Approximate Retention and Hibernation Characteristics
Table 4 - Hibernation vs. SoC Off Comparison
Table 5 - Approximate Beacon with SoC Off and Hibernation Characteristics
Table 6 - Approximate Active Flash and OTP Characteristics
Table A-1 Power Domains and Low Power States
Table A-2 Clock Gating Control and Low Power States

CONFIDENTIAL

Overview

The Atmosic ATM2/3 SoC supports active states and low power states as shown in Table 1 below. Each primary state may contain several secondary states as described in the ATM32xx and ATM22xx datasheets and Appendix A. Note that in some applications, "hibernate" state may be referred to as "sleep", while the "SoC Off" state may be called "deep sleep".

Table 1 - Power States

Primary Power State	Secondary Power State	PARAM_ID_SLEEP_ENABLE[1]	Description
Active	Active	0[2]	All regions of the SoC are powered on. Active power can be optimized by utilizing clock gating registers and/or by putting the Cortex-M0 into Wait for Interrupt (WFI).
	CPU Idle	0[2]	
	Bluetooth LE Deep Sleep	1[2]	
	Bluetooth LE Deep Sleep + CPU Idle	1[2]	
Retention	Retain Drop	2 or 3	Retain RAMs, flip-flops, I/Os states. Wakeup settings must be programmed before transitioning into this state. External flash is put into deep power down.
Hibernation	Hibernate	4	Powers down system memory. Retains minimal amount of flip-flop state. Retains I/O state. Wakeup settings must be programmed before transitioning into this state. External flash is put into deep power down.
SoC Off	SoC Off	5	All digital domains are powered down, but the PMU remains on in an ultra low-power state with limited functionality. External flash is put into deep power down.
Powerdown	PWD	Not Applicable	The PWD pin is asserted and all power domains including PMU are completely shut off

In beaconing applications, ATM2/3 transmits beacons periodically and remains idle in between those beacons. During the inactivity, ATM2/3 can be programmed into one of the three primary low power states: Retention, Hibernation or SoC Off to minimize the power consumption.

[1] NVDS tag, refer to the SDK User Reference Manual
[2] This setting is only used for debug

CONFIDENTIAL

In increasing order of steady state power consumption, the ranking is SoC Off then Hibernation then Retention. However, each mode comes with different restart overheads and programing flexibilities which make the best mode dependent on the use case scenarios.

Purpose

This document describes the functional and power consumption differences among the three modes and provides an in depth analysis of specific use cases. This document also describes the possible locations where the user application could reside and how that choice impacts the low power mode selection and operation.

Hibernation and Retention Comparison

In Retention state potentially all system memory state and all flip-flop state is retained. Hibernation state is an alternative to Retention state that achieves a lower power than Retention by powering down the system memory and by retaining only a minimal amount of flip-flop state. The time it takes to return from the selected low power state to an active state is different between Retention and Hibernation with Hibernation taking longer. Table 2 below compares the pros and cons between the two states.

Table 2 - Hibernation vs. Retention Comparison

	Hibernation	Retention
Steady state power consumption	Pro: Hibernation consumes less power because most digital logic circuits and all memory macros (ROM, OTP, RAM) are disconnected from power.	Con: Retention consumes more power because most digital logic circuits and all configured RAM macros are retained in a dormant state.
Restart overhead	Con: Waking from hibernation requires initialization that comes with a system reboot.	Pro: Waking from retention requires no initialization since the system continues from its retained state.
Data retention	Con: User data can only be retained in unused sensor hub registers or by overriding control signals for one of the RAM macros.	Pro: All user data in the RAM macros is automatically retained.
Wake up sources	Both modes offer the same wake up sources in keyboard detection, mouse movement detection, timer, wakeup receiver, and multiple GPIOs.	

CONFIDENTIAL

The need to preserve user data does not require selecting Retention over Hibernation. If the longer restart time is acceptable to the application and if the time between beacons is sufficiently long enough, the user should make every effort to pick Hibernation over Retention. One possibility is to store the user data in one of the 16 KB macros in system memory by setting the appropriate override control signals
in the power sequencer core (please refer to the RAM_hibernate SDK example). A second possibility is to store user data in unused sensor hub registers. With the first approach, up to 16 KB worth of user data can be stored (at the expense of one memory being on during Hibernation). With the second approach, up to 198 bytes can be stored.

Use Case Example

The ATM2/3 ROM contains a default Bluetooth beacon application with pre-configured parameters (e.g. beacon interval and low power state selection) that are customizable by NVDS tags in the OTP. In this use case, the ATM2/3 system transmits the beacon three times and then goes to one of the low power states for a programmable duration. The cycle repeats in this fashion indefinitely. The details on how to customize the OTP is described in the document: **SDK Quick Start Guide Configure Beacon Parameters in OTP** and the corresponding **README** files provided by the SDK. Table 3 shows the current consumption profile characteristics for both low power modes.

CONFIDENTIAL

Table 3 - Approximate Retention and Hibernation Characteristics

	Beacon with Retention	Beacon with Hibernation
Average low-power current I_{sleep} (µA)	2.3[3]	0.7
Average active/beacon current[4] I_{active} (µA)	700	600
Total time of active/beacon T_{active} (ms)	6.2	14
Time from wakeup to beacon (ms)	3.1	11

Note that hibernation has a lower current floor but takes longer time to restart beaconing.

[3] I_{sleep} for Beacon with Retention is 2.6 µA for ATMx2xx-x0x. Refer to ATMx2xx Chip Revisions Application Note for information regarding chip revisions.
[4] All measurements are conducted at 0 dBm transmit power

Power as a function of increasing beacon interval for both cases was computed using the equations:

$$E_{active} = I_{active} \cdot V_{batt} \cdot T_{active}$$

$$P(t) = \frac{E_{active} + I_{sleep} \cdot V_{batt} \cdot t}{T_{active} + t}$$

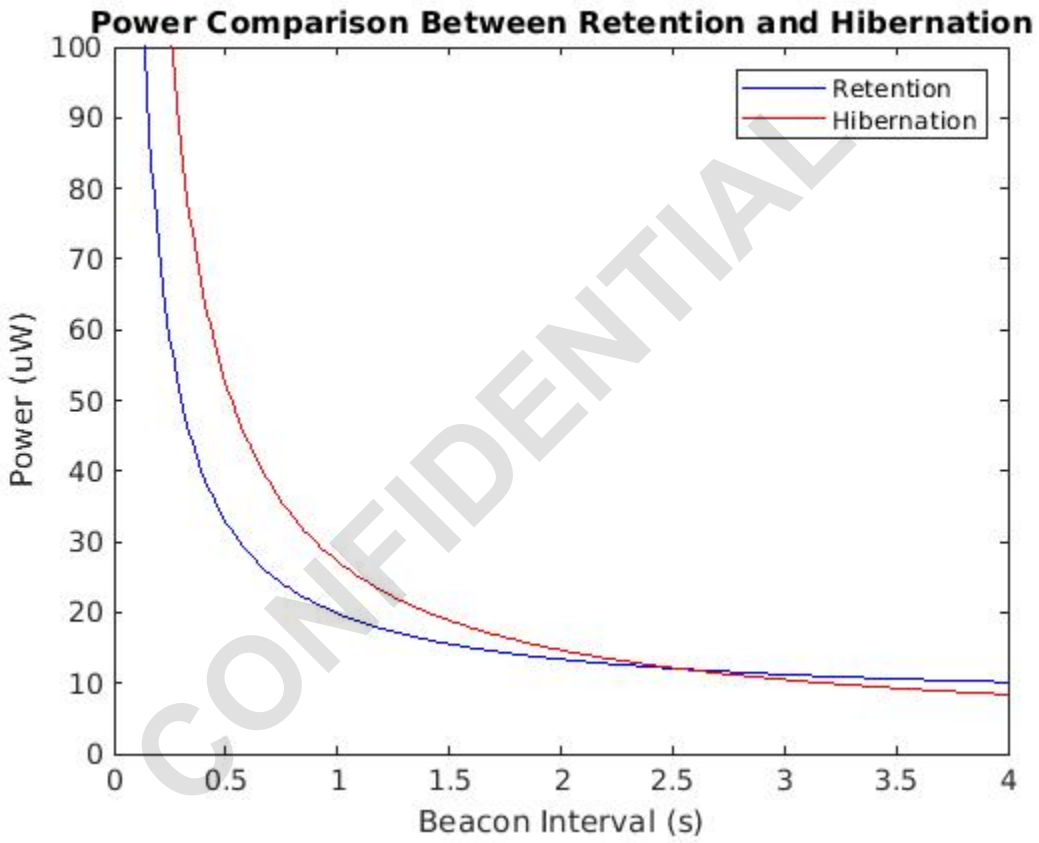


Figure 1 - Power Comparison Between Retention and Hibernation

Based on Figure 1, Retention is more power efficient for duration of < 2.5 s and hibernation becomes more efficient beyond that.

Hibernation and SoC Off Comparison

All state elements are reset upon coming out of either hibernation and SoC Off modes, therefore the SoC system reboots upon coming out of either mode. Table 4 below shows the pros and cons between the two modes:

CONFIDENTIAL

Table 4 - Hibernation vs. SoC Off Comparison

	Hibernation	SoC Off
Steady state power consumption	Con: Hibernation consumes more power because PMU supplies are not shut off and some digital logic circuits are still active.	Pro: SoC Off consumes less power because some PMU supplies are shut off and only certain PMU logic circuits are still active.
Restart overhead	Pro: Waking from hibernation only requires initialization that comes with a system reboot.	Con: Waking from SoC Off requires initialization that comes with a system reboot as well as significant inrush current that comes with restoring PMU supplies.
Data retention	Pro: User data can be retained in unused sensor hub registers or by overriding control signals for one of the RAM macros.	Con: No user data can be retained.
Wake up sources	Pro: Wake up sources include keyboard detection, mouse movement detection, timer, wakeup receiver, and multiple GPIOs.	Con: SoC Off has a smaller subset of wakeup sources which include a timer, a single GPIO trigger, and a low power analog comparator.

CONFIDENTIAL

Table 5 shows the current consumption profile characteristics for both low power modes.

Table 5 - Approximate Beacon with SoC Off and Hibernation Characteristics

	Beacon with SoC Off	Beacon with Hibernation
Average low-power current I_{sleep} (µA)	0.30	0.7
Average active/beacon current I_{active} (µA)	700[5]	600
Total time of active/beacon T_{active} (ms)	17	14
Time from wakeup to beacon (ms)	14	11

Power as a function of increasing beacon interval for both cases was computed using the equations:

$$E_{active} = I_{active} \cdot V_{batt} \cdot T_{active}$$

$$P(t) = \frac{E_{active} + I_{sleep} \cdot V_{batt} \cdot t}{T_{active} + t}$$

CONFIDENTIAL

[5] Active current for Beacon/SoC Off is higher than for Beacon/Hibernation because returning from SoC Off requires that more power domains be charged up. All charging current is included in this average. In practice some of this charging current will come not from the battery but from the board decoupling capacitors. Ignoring this charging current, the initialization sequence for both SoC Off and Hibernation are largely the same.

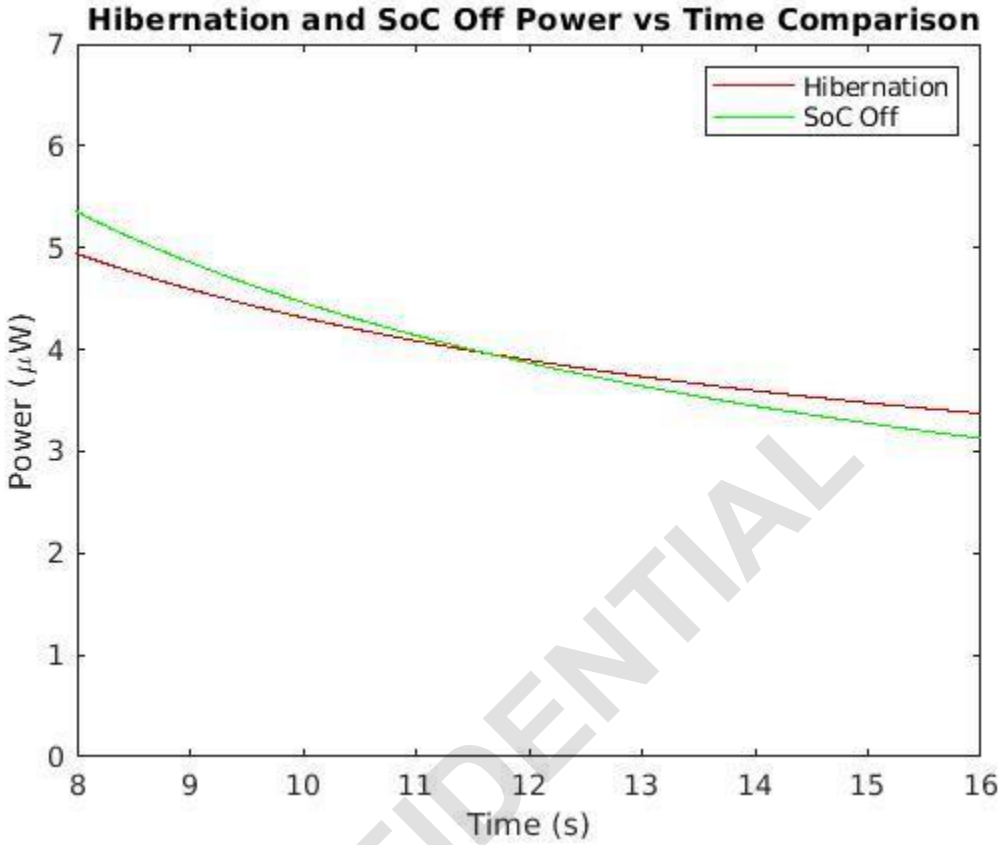


Figure 2 - Hibernation and SoC Off Average Power vs Time Comparison

From Figure 2, Hibernation is more efficient for interval of < 12 s and SoC Off becomes more efficient beyond that point.

Summary:

Comparison between Hibernation and Retention, Retention is more efficient when time interval of inactivity is < 2.2 s

Comparison between SoC Off and Hibernation, Hibernation is more efficient when time interval of inactivity is < 12 s

CONFIDENTIAL

Program Storage and Low Power State

In this section, the process of deciding where a user application should be stored and the choice of low power state to optimize for power efficiency is discussed.

Developing a User-Defined Program

For applications that are not supported by the ROM, the application code must be executed from either flash, OTP, or RAM. Figure 3 below illustrates a process of developing a user-defined cyclical application with the goal for optimal power consumption. The cyclical nature of the program means each interval starts with an active portion and followed by an inactive duration. The choice of low power state used in the inactive duration impacts the restart overhead of each interval in time and energy and determines how low the power consumption is during inactivity.

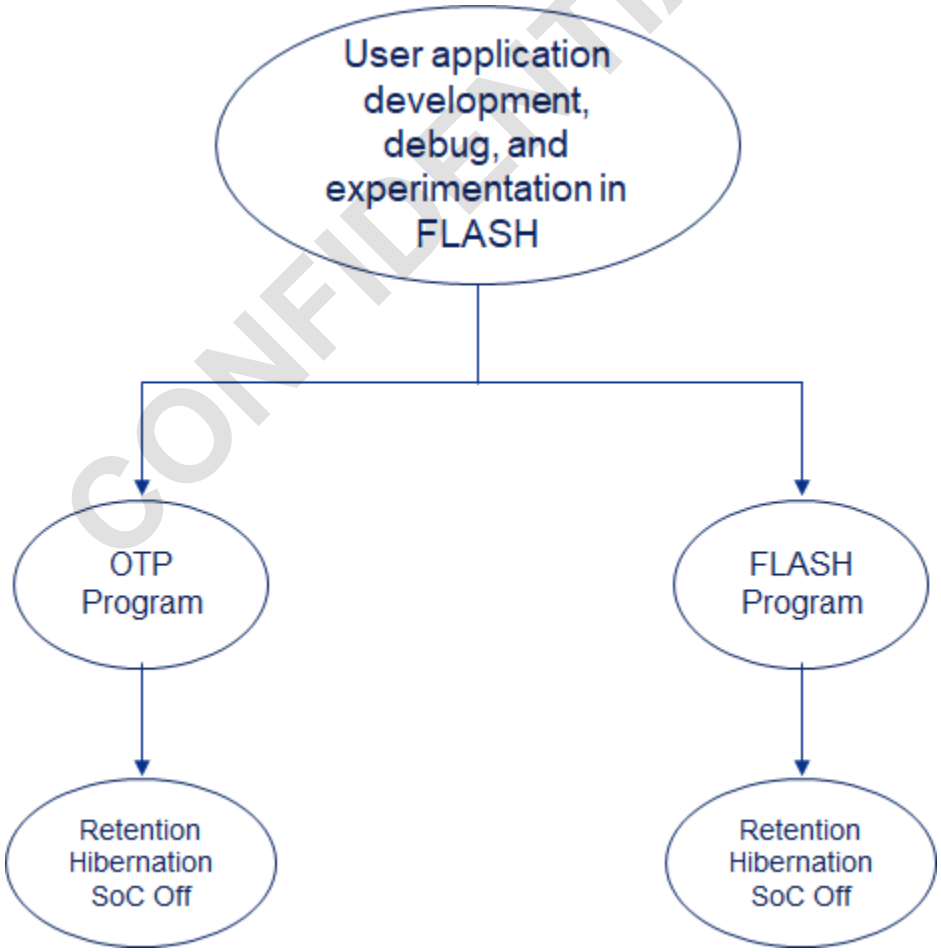


Figure 3 - User-defined Cyclical Application Development Process

In the development phase, the flash offers the most space for debugging and experimentation. The final program can be stored in one of these locations:

CONFIDENTIAL

1. Flash: offers the largest space (up to 16 MB of external flash capacity) and supports all low power modes; however, flash accesses consume more power than accesses to OTP. Two modes of execution are supported to mitigate the higher energy cost.
    a. Flash + QSPI cache enabled: the cache significantly reduces the number of flash accesses during execution and is retained during Retention. In SoC Off or Hibernation, the cache is invalidated.
    b. Flash + RAM download: The program is downloaded to RAM from flash at initialization. Flash is no longer accessed after the initial loading but the total space of RAM (including data memory) is only 96 KB. If the low power state is Retention, then the program can cycle between active and retention perpetually without flash. However, if SoC Off or Hibernation is chosen as the low power state, then the initial loading is always required prior to the active state, resulting in longer boot time than running with flash + cache enabled.
2. OTP: offers limited program space (4 KB) and supports all low power modes. For programs that can fit and do not need to be updated, this option is lower power than flash. The program is downloaded to RAM from OTP at initialization. If the low power state is Retention, then the program can cycle between active and retention perpetually without accessing the OTP. However, if SoC Off or Hibernation is chosen as the low power state, then the initial loading is always required prior to the active state, resulting in longer boot time than running with flash + cache enabled.

CONFIDENTIAL

Examination of SDK's tmp1075_sensor_adv Application

The tmp1075_sensor_adv example in the SDK involves polling temperature data from an external sensor via the I2C protocol and beaconing the data off to an external host.
This application was picked for examination because it was designed to be small enough to fit in OTP and can be easily modified to support Retention and avoid cache misses.
Table 6 compares the active current consumption profile characteristics for each beacon interval with the program code stored in flash vs OTP.

Table 6 - Approximate Active Flash and OTP Characteristics

	Flash + Cache Hibernation	Flash + Cache Retention	Flash + RAM Hibernation	Flash + RAM Retention	OTP Hibernation
Average Active Current I_{active} (µA)	900	750	800	750	550
Total Active Time (ms)	20	10	20.5	10	21

Note that the user may also use both Hibernation and Retention in the same application in which long activity intervals are interspersed among bursts of short intervals. It may make sense to use Hibernation for long inactivities and Retention for short ones.

CONFIDENTIAL

Appendix A

This appendix provides more details on how the power domains and control gating affect each power state.

Table A-1 Power Domains and Low Power States

Primary Power State	Secondary Power State	PARAM_ID_SLEEP_ENABLE	Power Domains							
			Bluetooth LE (RF, Link Controller, Modem)	CPU (M0+Peripheral -RF)	Efuse	System ROM	System RAM 96 KB	Exchange Memory 32 KB	Top 32 kHz	PMU
Active	Active	0	On	On	On	On	On	On	On	On
	CPU Idle	0	Off	On	On	On	On	On	On	On
	Bluetooth LE Deep Sleep	1	Off	On	On	On	On	On	On	On
	Bluetooth LE Deep Sleep + CPU Idle	1	Off	On	On	On	On	On	On	On
Retention	Retain_Drop	2 or 3	Registers retained	Registers retained	Off	Off	16-96 KB retained*	16-32 KB retained*	On	On
Hibernation	Hibernate	4	Off	Off	Off	Off	Off	Off	On	On
SoC Off	SoC Off	5	Off	Off	Off	Off	Off	Off	Off	Partially On
Powerdown	PWD	Not Applicable	Off	Off	Off	Off	Off	Off	Off	Off

*VDD lowered

CONFIDENTIAL

Table A-2 Clock Gating Control and Low Power States

Primary Power State	Secondary Power State	PARAM_ID_SLEEP_ENABLE	SW Clock Gating Control					
			16 MHz to Bluetooth LE	16 MHz to CPU	32 KHz	IP Core	WURX	GADC
Active	Active	0	On	On	On	Yes	Yes	Yes
	CPU Idle	0	On	WFI	On	Yes	Yes	Yes
	Bluetooth LE Deep Sleep	1	Turn Off	On	On	Yes	Yes	Yes
	Bluetooth LE Deep Sleep + CPU Idle	1	Turn Off	WFI	On	Yes	Yes	Yes
Retention	Retain_Drop	2 or 3	PSEQ Off	PSEQ Off	On	PSEQ Off	Yes	Off
Hibernation	Hibernate	4	PSEQ Off	PSEQ Off	On	PSEQ Off	Yes	Off
SoC Off	SoC Off	5	Off	Off	Off	Off	Off	Off
Powerdown	PWD	Not Applicable	Off	Off	Off	Off	Off	Off

CONFIDENTIAL



CONFIDENTIAL

ATMOSIC TECHNOLOGIES - DISCLAIMER

This product document is intended to be a general informational aid and not a substitute for any literature or labeling accompanying your purchase of the Atmosic product. Atmosic reserves the right to amend its product literature at any time without notice and for any reason, including to improve product design or function. While Atmosic strives to make its documents accurate and current, Atmosic makes no warranty or representation that the information contained in this document is completely accurate, and Atmosic hereby disclaims (i) any and all liability for any errors or inaccuracies contained in any document or in any other product literature and any damages or lost profits resulting therefrom; (ii) any and all liability and responsibility for any action you take or fail to take based on the information contained in this document; and (iii) any and all implied warranties which may attach to this document, including warranties of fitness for particular purpose, non-infringement and merchantability. Consequently, you assume all risk in your use of this document, the Atmosic product, and in any action you take or fail to take based upon the information in this document. Any statements in this document in regard to the suitability of an Atmosic product for certain types of applications are based on Atmosic's general knowledge of typical requirements in generic applications and are not binding statements about the suitability of Atmosic products for any particular application. It is your responsibility as the customer to validate that a particular Atmosic product is suitable for use in a particular application. All content in this document is proprietary, copyrighted, and owned or licensed by Atmosic, and any unauthorized use of content or trademarks contained herein is strictly prohibited.

Copyright ©2019-2021 by Atmosic Technologies. All rights reserved.

www.atmosic.com